

Hierarchical Clustering

Importing the libraries

```
In [3]: import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

Importing the dataset

```
In [8]: dataset = pd.read_csv(r"C:\Users\sande\OneDrive\Desktop\Naresh IT\data science and dataset")
```

```
Out[8]:
```

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)	k-means
0	1	Male	19	15	39	NaN
1	2	Male	21	15	81	NaN
2	3	Female	20	16	6	NaN
3	4	Female	23	16	77	NaN
4	5	Female	31	17	40	NaN
...	...	...	...	...	...	...
195	196	Female	35	120	79	NaN
196	197	Female	45	126	28	NaN
197	198	Male	32	126	74	NaN
198	199	Male	32	137	18	NaN
199	200	Male	30	137	83	NaN

200 rows × 6 columns

```
In [9]: x = dataset.iloc[:, [3, 4]].values
```

Using the dendrogram to find the optimal number of clusters

```
In [ ]: import scipy.cluster.hierarchy as sch
dendrogram = sch.dendrogram(sch.linkage(x, method = 'ward'))
plt.title('Dendrogram')
plt.xlabel('Customers')
```

```
plt.ylabel('Euclidean Distances')
plt.show()
```

Training the Hierarchical Clustering model on the dataset

```
In [11]: from sklearn.cluster import AgglomerativeClustering
hc = AgglomerativeClustering(n_clusters = 5, metric = 'euclidean', linkage = 'ward')
y_hc = hc.fit_predict(x)
```

Visualising the clusters

```
In [12]: plt.scatter(x[y_hc == 0, 0], x[y_hc == 0, 1], s = 100, c = 'red', label = 'Cluster 1')
plt.scatter(x[y_hc == 1, 0], x[y_hc == 1, 1], s = 100, c = 'blue', label = 'Cluster 2')
plt.scatter(x[y_hc == 2, 0], x[y_hc == 2, 1], s = 100, c = 'green', label = 'Cluster 3')
plt.scatter(x[y_hc == 3, 0], x[y_hc == 3, 1], s = 100, c = 'cyan', label = 'Cluster 4')
plt.scatter(x[y_hc == 4, 0], x[y_hc == 4, 1], s = 100, c = 'magenta', label = 'Cluster 5')
plt.title('Clusters of customers')
plt.xlabel('Annual Income (k$)')
plt.ylabel('Spending Score (1-100)')
plt.legend()
plt.show()
```

